🛒 E-Commerce Backend Project Documentation

1. 📌 Project Title

E-Commerce Backend Application using Spring Boot

2. 🎯 Objective

The objective of this project is to develop a backend system for an e-commerce application that allows users to register, login, browse products, add items to cart, and manage orders securely using REST APIs.

3. 🛠️ Technologies Used

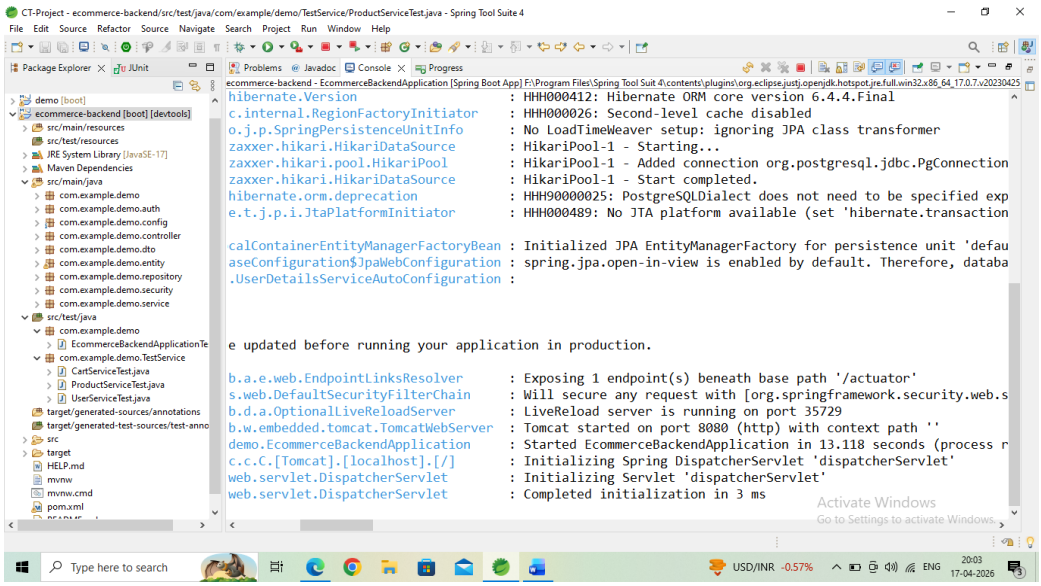
- Programming Language: Java
- Framework: Spring Boot
- Database: PostgreSQL
- ORM Tool: Hibernate (JPA)
- Security: Spring Security + JWT Authentication
- Build Tool: Maven
- Server: Embedded Tomcat
- API Testing: Postman

4. 🏗️ Project Architecture

The project follows Layered Architecture:

- Controller Layer → Handles HTTP requests
- Service Layer → Business logic
- Repository Layer → Database interaction

- Entity Layer → Database tables



5. 🔑 Features

👤 User Management

- User Registration
- User Login with JWT Token
- Role-based authentication

📦 Product Management

- Add Product
- View Products
- Update/Delete Product (Admin)

🛒 Cart Management

- Add to Cart
- View Cart
- Remove from Cart

📦 Order Management

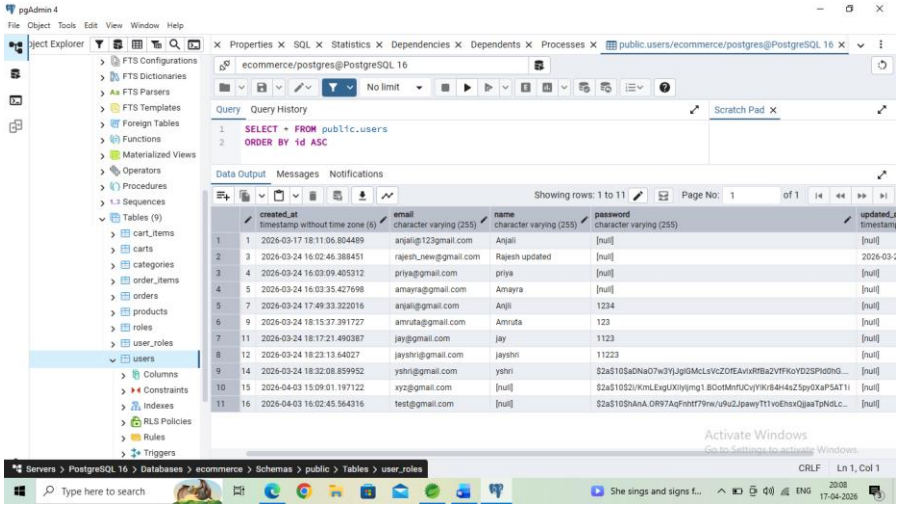
- Place Order
- View Orders

6. 🗂️ Database Design

Tables:

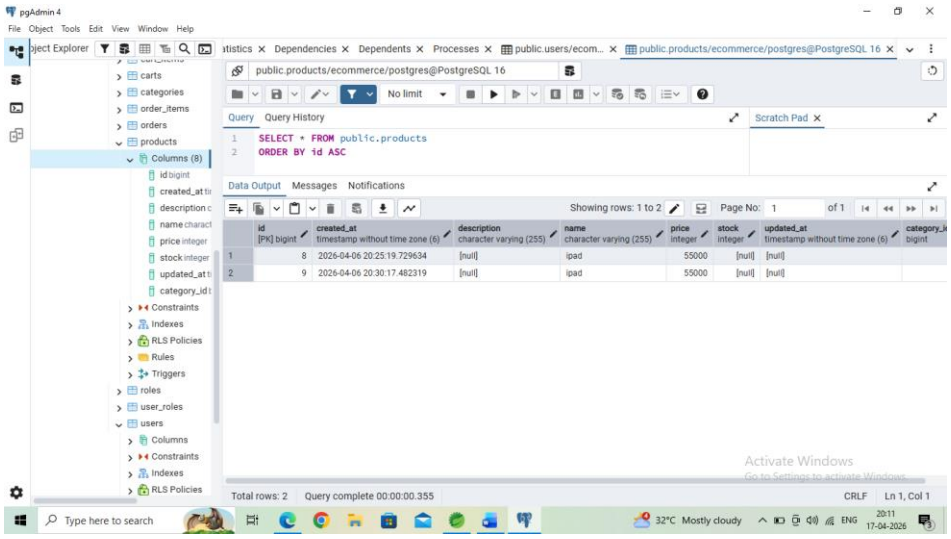
👤 User Table

- id
- name
- email
- password
- role



📦 Product Table

- id
- name
- price
- description



🛒 Cart Table

- id
- user_id
- product_id
- quantity

📄 Order Table

- id
- user_id
- total_amount
- order_date

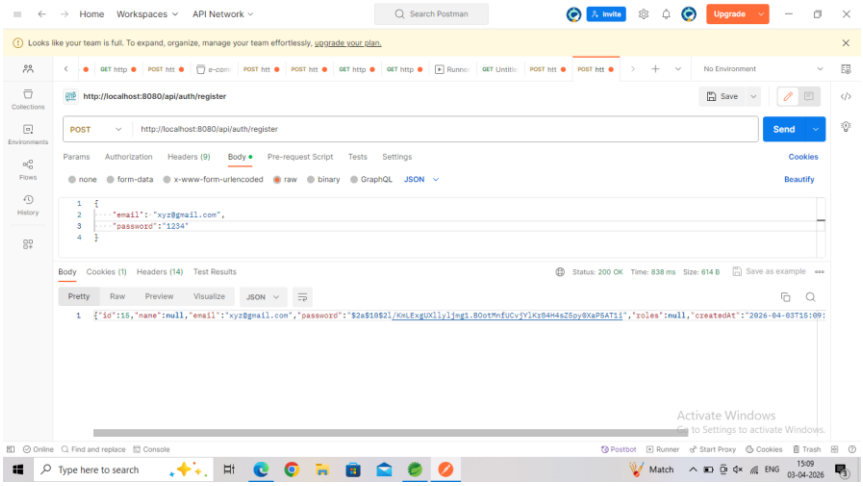
7. 🔐 Security Implementation

- Used JWT (JSON Web Token) for authentication
- Password encrypted using BCrypt
- Secured APIs using Spring Security Filters
- Unauthorized access returns 403 Forbidden

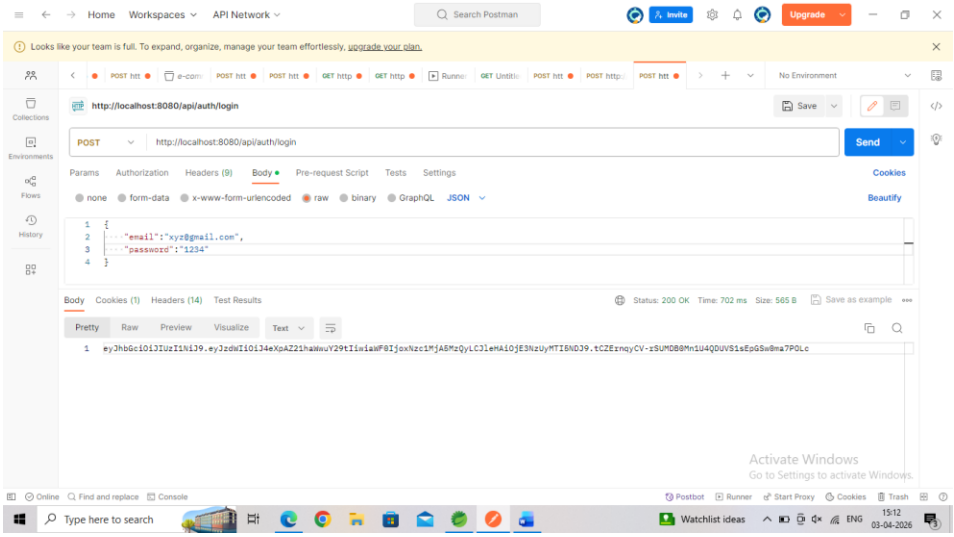
8. 🔄 API Endpoints

Auth APIs

- POST /api/auth/register → Register user

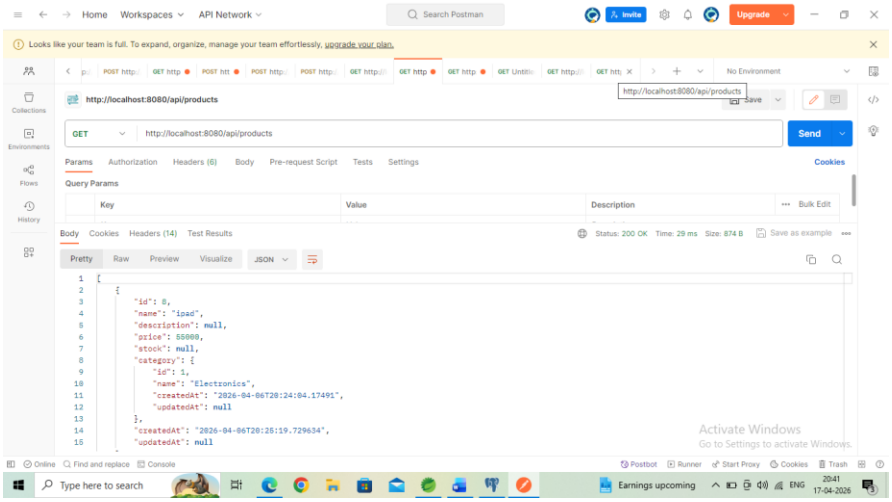


- POST /api/auth/login → Login and get token

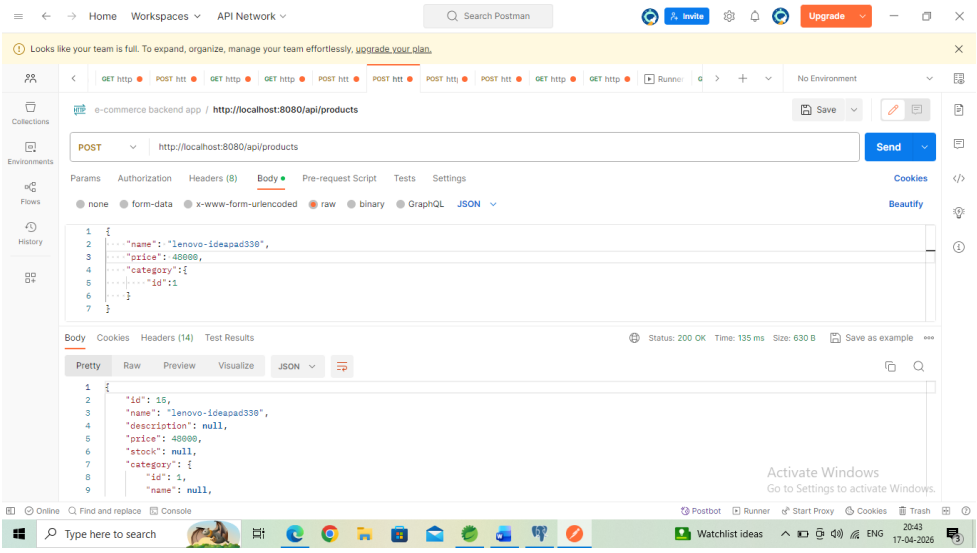


Product APIs

- GET /api/products → Get all products

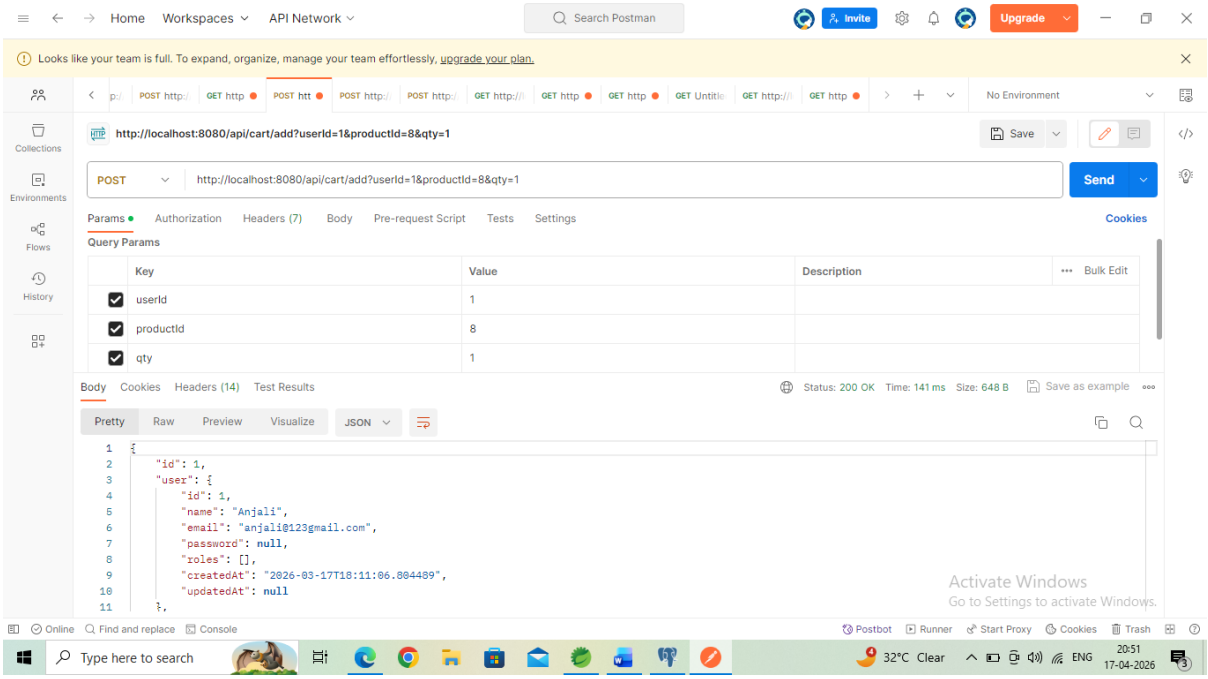


- POST /api/products → Add product

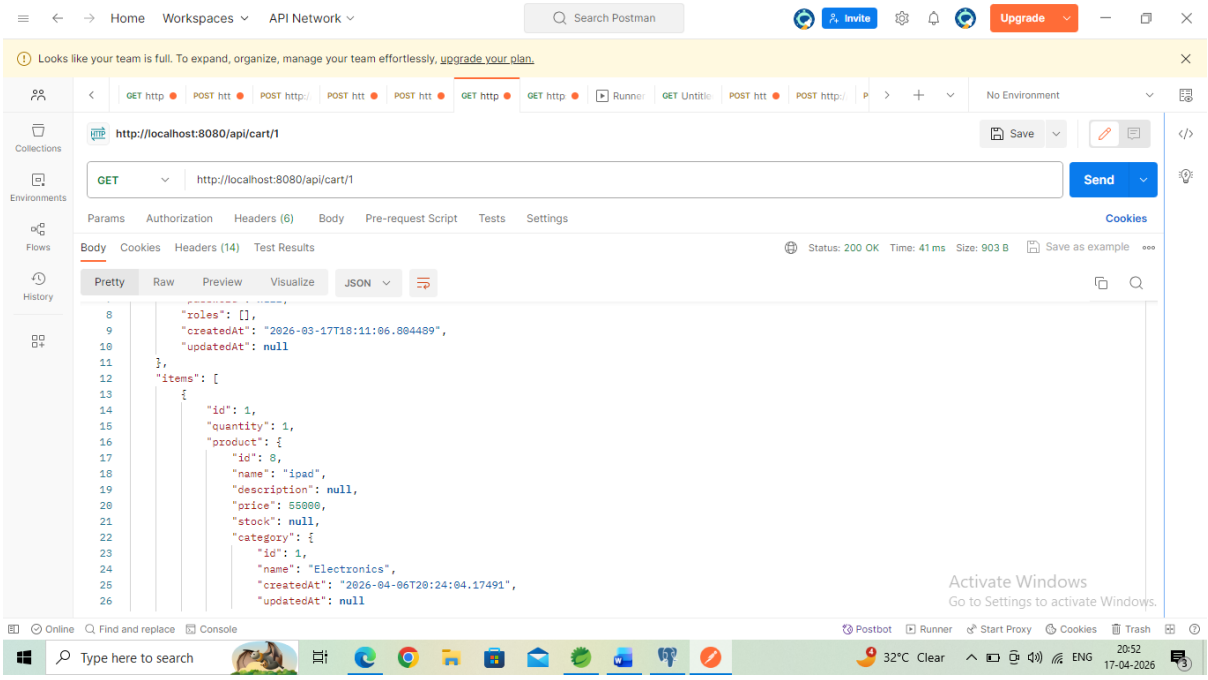


Cart APIs

- POST /api/cart/add → Add item to cart



- GET /api/cart → View cart



Order APIs

- POST /api/orders → Place order
- GET /api/orders → View orders

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9. ⚙️ How to Run the Project

1. Clone the repository
2. Open in **Spring Tool Suite (STS)** or IntelliJ
3. Configure PostgreSQL database in application.properties
4. Run the application
5. Use Postman to test APIs

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10. 🚧 Challenges Faced

- Handling JWT authentication errors
- Managing relationships between entities
- Fixing 403 Forbidden error in security configuration

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11. 📈 Future Enhancements

- Add Payment Gateway Integration
- Add Frontend (React / Angular)
- Add Product Images Upload
- Add Admin Dashboard

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12. 📌 Conclusion

This project demonstrates the implementation of a secure and scalable backend system for an e-commerce application using Spring Boot. It covers real-world concepts like authentication, REST APIs, and database management.

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13. 🧑‍💻 Developed By

Anjali Nirbhawane

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